

EEE F435 Digital Image Processing Assignment-II

This assignment will enhance your understanding of the role of the Fourier Transform in image filtering using MATLAB. Perform various operations on an image to analyse how different frequency-domain filters affect the visual representation of the image.

Input Image for all the examples:

Input images mean (Image 1 and Image 2, both)

Image 1: An image captured for assignment 1.

Image 2: An image of you with the instructor in the classroom with a dynamic background during the lecture hour.

Use the values of A and B as mentioned below:

A=4th digit of BITS ID; B=10th digit of BITS ID;

Example: BITS ID: 2024AA1234P; A=4 and B=4

For the property-based questions, always compare the image and spectrum with the original.

Submission Requirements

A single PDF file containing

- Student and course details
- MATLAB code (.m file) implementation.
- Screenshots of **original and processed images** for each filter.
- A short report including:
 - Explanation of each filter.
 - Compare different filters and their effects on images.
 - Conclusion: When would you prefer to use different frequency filters? Please provide an example for each filter.

Submission Deadline: 5:00 pm, Wednesday, 1 October 2025

Task I: General Understanding

Write a MATLAB script that performs the following operations on both the images (Image 1 and Image 2). Include relevant comments in your code to explain each step taken.

1. Display the original images, respective magnitude spectrum (log scale), and phase spectrum.
2. Why is `fftshift` required in MATLAB? Justify with proper experimentation.
3. Swap the magnitude of images with the phase of another and reconstruct. Comment on what this experiment proves.
4. Shifting Property: Shift the input images by $(A*10+5, B*10+5)$ pixels. Show effect on magnitude and phase spectrum.
5. Scaling Property: Compress the image by a factor of A and expand the image by a factor of B. Show the effect on its frequency spread.
6. Rotation Property: Rotate images by $30^\circ, 45^\circ, 60^\circ, 90^\circ$. Show how the spectrum rotates.

Task-II: Filters

- Load the input images and apply the following Filter over the images.
 - Output the original images and the filtered images.
 - Plot the magnitude and phase of both the original and the filtered images.
 - **Compare the effect of filter order size on the level of blur, edge preservation, and ringing artifacts.** Comment on your observation
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1. Apply an Ideal (Disk and Rectangle) Low Pass Filter,
 2. Butterworth (orders 1, 2, 4) Low Pass Filter,
 3. Gaussian LPF with cutoffs ($D_0=25, 50, 100$)
 4. Apply an Ideal (Disk and Rectangle) High Pass Filter,
 5. Butterworth (orders 1, 2, 4) High Pass Filter,
 6. Gaussian HPF with cutoffs ($D_0=30, 60, 120$).
 7. Add synthetic sinusoidal noise of frequency $(A*10+100)$ to both the input images.
 8. Remove it using a suitable cutoff frequency Band-Reject.
 9. **Compare the different LPF/HPF/BR filters and** comment on your observation